Multiple-choice questions

|!|BCEMC0250000001|!|

What is the function of plasma cells in an immune response?

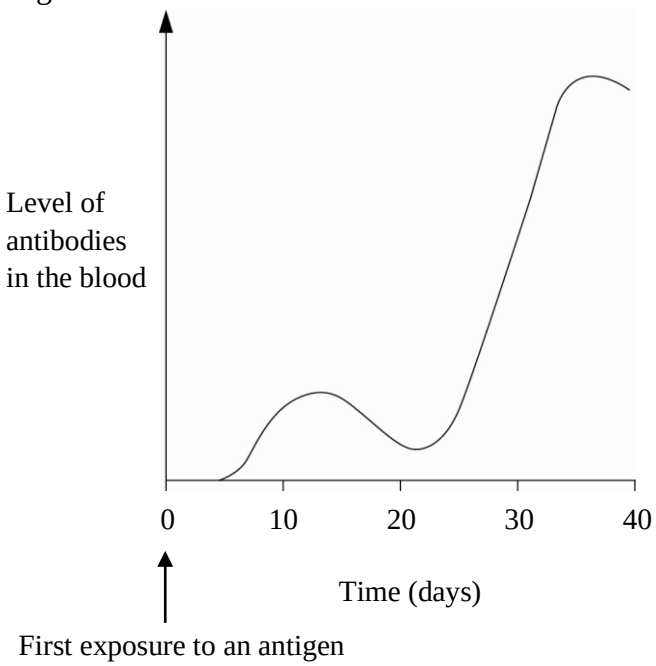
A. to engulf bacteria
B. to produce antibodies
C. to develop memory B cell
D. to destroy cells infected with virus

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0250000002|!| ★

The following graph shows the level of antibodies in the blood of a person who has been infected with a pathogen.



Which of the following statements about the graph is correct?

A. Antibodies were injected on day 5.
B. Helper T cells were activated on day 12.
C. It took 35 days to achieve active immunity.
D. Memory cells for this pathogen lead to the surge in antibody level on around day 22.

##D##

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Skill assessed	Mark
Analysing data	1

|!|BCEMC0250000003|!|

Which of the following is ***not*** a chemical barrier that kills or inhibits the growth of pathogens?

A. sebum
B. tear
C. lymph
D. gastric juice

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000004|!|

Which of the following is ***not*** an action of antibodies?

A. making holes on the surfaces of pathogens
B. killing pathogens by phagocytosis
C. neutralizing the toxins secreted by pathogens
D. causing pathogens to clump together to prevent them from reproducing

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000005|!|

Which of the following does ***not*** belong to non-specific defence mechanisms?

A. mucous membranes of the respiratory tract
B. gastric juice
C. plasma cells
D. skin

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0250000006|!|

Which of the following belong(s) to non-specific defence mechanisms?

(1) blood clotting
(2) secretion of gastric juice
(3) phagocytosis

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000007|!|

Which of the following is ***not*** a chemical barrier?

A. tear
B. sebum
C. mucus secreted by the lining of the respiratory tract
D. gastric juice

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000008|!|

Which of the following organs produce(s) secretions with a pH lower than 7?

(1) vagina
(2) small intestine
(3) stomach

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

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##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0250000009|!| ★

Which of the following combinations about phagocytes and lymphocytes is ***incorrect***?

	Phagocytes	***Lymphocytes***
A.	belong to the non-specific defence mechanism	belong to the specific defence mechanism
B.	involved in inflammatory responses	not involved in inflammatory responses
C.	produced in the bone marrow	produced in the thymus
D.	have a lobed nucleus	have a round nucleus

☐

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000010|!|

Which of the following statements about antibodies is ***incorrect***?

A. Antibodies are produced by plasma cells.
B. The production of antibodies belongs to non-specific defence mechanism.
C. Antibodies are not re-usable.
D. Antibodies are Y-shaped protein molecules.

☐

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000011|!|

Which of the following cells is/are differentiated from a B cell?

(1) T cell
(2) memory B cell
(3) plasma cell

A. (2) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

☐

##D##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000012|!|

Which of the following statements about the primary response is correct?

A. The primary response is induced by the first entry of an antibody.
B. Disease symptoms do not appear during the primary response.
C. During the primary response, no antibodies are produced against the pathogens.
D. Memory cells are produced during the primary response.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000013|!|

Which of the following statements about different types of T cells is/are correct?

(1) Killer T cells produce antibodies to act against pathogens.
(2) Helper T cells release lymphokines to activate phagocytes.
(3) Memory T cells ‘memorize’ the pathogens which have invaded the body.

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0250000014|!|

Which of the following defence mechanisms are non-specific?

(1) inflammatory response
(2) production of antibodies
(3) phagocytosis

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000015|!|

Antigens are present on the surfaces of

(1) bacteria.
(2) red blood cells.
(3) viruses.

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000016|!|

Which of the following combinations ***incorrectly*** matches the cell involved in body defence with its function?

	Cell	***Function***
A.	memory B cell	releases lymphokines
B.	mast cell	releases histamine
C.	plasma cell	produces antibodies
D.	killer T cell	destroys infected cells directly

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##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0250000017|!| ★

Which of the following is an artificial way to acquire active immunity?

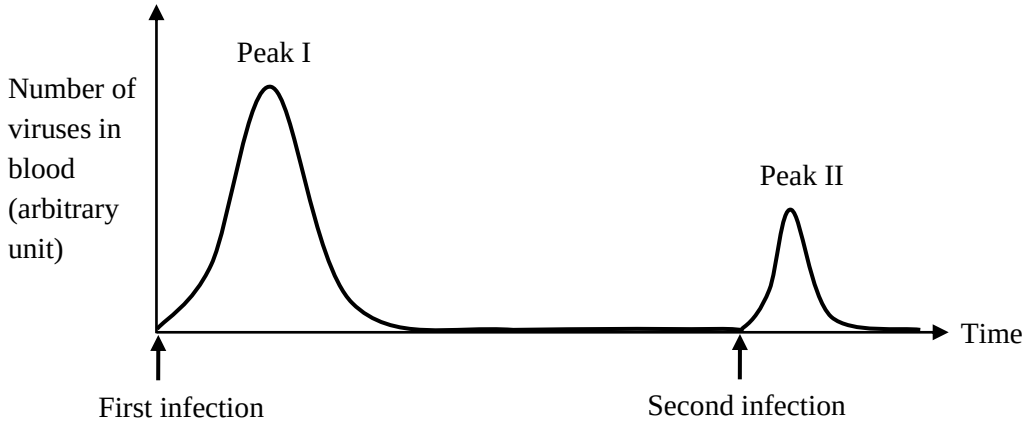
A. recovering from a disease
B. vaccination
C. injection of serum
D. breast feeding

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0250000018|!| ★

The graph below shows the changes in the number of viruses in the blood of a person after the first and second infections of a virus.



Which of the following statements correctly explain(s) why Peak II is lower than Peak I?

(1) The plasma cells produce a large amount of antibodies within a short time during second infection.
(2) A large amount of killer T cells are formed to destroy infected cells during second infection.
(3) More memory cells are formed during second infection.

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A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0250000019|!|

Which of the following blood components can destroy bacteria?

(1) antibody
(2) blood platelet
(3) phagocyte

A. (1) only
B. (2) only
C. (1) and (2) only
D. (1) and (3) only

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000020|!|

Which of the following statements about antibodies is/are correct?

(1) Antibodies are specific to certain antigens.
(2) Antibodies are produced by T cells.
(3) Antibodies can multiply upon the attack of bacteria.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0250000021|!|

Which of the following statements about chemical barriers against pathogens is ***incorrect***?

A. Acidic secretion produced by the vagina can inhibit the growth of pathogens.
B. Tears contain white blood cells which can kill pathogens.
C. Gastric juice contains hydrochloric acid which can kill pathogens.
D. Sebum secreted by the sebaceous gland contains chemicals that kill certain pathogens.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000022|!|

Which of the following are the actions of antibodies against pathogens?

(1) making holes on the surfaces of pathogens
(2) disrupting cell wall formation in bacteria
(3) causing pathogens to clump together

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0250000023|!|

Which of the following secretions contain(s) lysozyme?

(1) tear
(2) gastric juice
(3) saliva

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000024|!|

Which of the following substances is/are essential for blood clotting?

(1) fibrinogen
(2) histamine
(3) platelets

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0250000025|!|

Which of the following responses is/are antigen-specific?

(1) phagocytosis
(2) immune response produced by T cells
(3) inflammatory response

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000026|!|

Which of the following is ***not*** a physical barrier in body defence against pathogens?

A. skin
B. hair in the nostrils
C. ciliated epithelium of the respiratory tract
D. gastric gland

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000027|!|

Which of the following is/are involved in the inflammatory response?

(1) phagocyte
(2) histamine
(3) lymphocyte

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

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##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000028|!|

Which of the following structures of the skin can defend against pathogens?

(1) sebaceous gland
(2) the outermost layer of the epidermis
(3) subcutaneous fat

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000029|!|

Which of the following components of blood ***cannot*** destroy pathogens?

A. killer T cells
B. antibodies
C. blood platelets
D. phagocytes

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000030|!|

T cells mature in

A. the thymus gland.
B. the bone marrow.
C. the lymph node.
D. the liver.

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##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000031|!|

How do phagocytes protect the human body against pathogens?

(1) They seal the wound to prevent the entry of pathogens into the human body.
(2) They neutralize the toxins produced by pathogens.
(3) They engulf and digest pathogens.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000032|!|

Which of the following substances cause(s) the dilation of arterioles near a wound?

(1) histamine
(2) lymphokine
(3) fibrinogen

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0250000033|!|

Which of the following statements about the inflammatory response is ***incorrect***?

A. Tissue fluid accumulates at the wound.
B. More mast cells are squeezed through the blood capillary wall into the tissue fluid.
C. More phagocytes engulf the invading pathogens.
D. The blood flow to the wound increases.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000034|!|

Which of the following statements about antibodies is ***incorrect***?

A. They are proteins.
B. They are produced by plasma cells.
C. They have memory for specific antigens.
D. They are transported in blood to different parts of the body.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000035|!|

Which of the following is/are the action(s) of antibodies on pathogens?

(1) inhibiting protein synthesis in pathogens
(2) causing pathogens to lyse
(3) causing pathogens to clump together

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0250000036|!|

Which of the following combinations ***incorrectly*** matches the blood cell with its function?

	Blood cell	***Function***
A.	red blood cell	carrying oxygen
B.	lymphocyte	non-specific body defence
C.	phagocyte	engulfing pathogens
D.	blood platelet	blood clotting

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000037|!|

Which of the following statements about antigens is ***incorrect***?

A. Antigens can be found on the surfaces of red blood cells.
B. All antigens are proteins.
C. Antigens can bind to specific antibodies.
D. Antigens can stimulate T cells to develop into plasma cells.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000038|!|

Which of the following occur(s) in an inflammatory response?

(1) The permeability of capillary walls increases, allowing more phagocytes to reach the damaged tissue.
(2) More tissue fluid accumulates in the wound.
(3) The arterioles near the wound constrict to reduce blood loss from the wound.

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

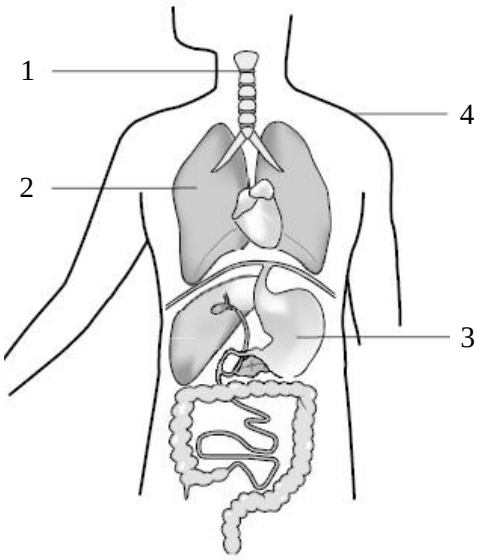
##B##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000039|!|

Directions: The following two questions refer to the diagram below which shows a part of the human body.



Which of the following physical barriers is/are present in structure 1?

(1) hair
(2) mucus
(3) closely packed epithelial cells

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0250000040|!|

Which of the following combinations about the secretions produced by the labelled structures for body defence is correct?

	Structure 1	*Structure 3*	*Structure 4*
A.	saliva	gastric juice	sweat
B.	mucus	gastric juice	sebum
C.	saliva	mucus	sebum
D.	mucus	mucus	sweat

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0250000041|!|

Which of the following is/are the importance of blood clotting in humans?

(1) preventing the entry of pathogens
(2) preventing excessive blood loss
(3) stimulating antibody production

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0250000042|!|

Which of the following statements about the cells involved in body defence is/are correct?

(1) Memory B cells can differentiate into plasma cells.
(2) Helper T cells can release lymphokines.
(3) Killer T cells can destroy cancer cells directly.

A. (3) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3) only

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000043|!|

Which of the following cells can produce antibodies?

A. *Penicillium*
B. plasma cell
C. phagocyte
D. T cell

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000044|!|

Which of the following statements about antigens is ***incorrect***?

A. They can be found on the surface of pathogens.
B. They can stimulate the production of antibodies in humans.
C. They cannot be found on the surface of human cells.
D. They can bind to specific antibodies.

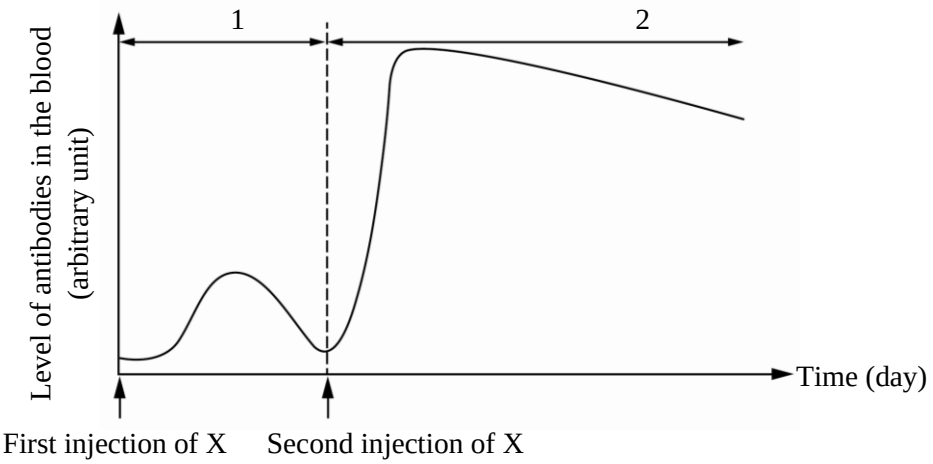
##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0250000045|!| ★

Directions: The following two questions refer to the graph below, which shows the changes in the level of antibodies in the blood.



What can X be?

(1) vaccine
(2) antibodies
(3) pathogens

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0250000046|!| ★

Which of the following comparisons between period 1 and period 2 is/are correct?

(1) More antibodies are produced in period 2.
(2) It takes a longer time for the antibodies to be produced in period 1.
(3) No memory cells are produced in period 2.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

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##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0250000047|!|

A vaccine may contain

A. antibiotics.

B. antibodies.

C. killed or weakened pathogens.

D. active pathogens.

□

##C##

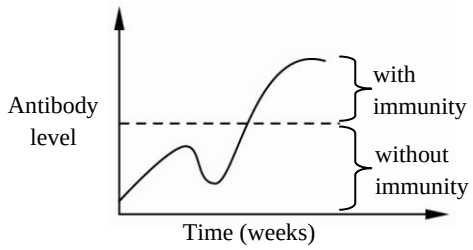
Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0250000048|!| ★

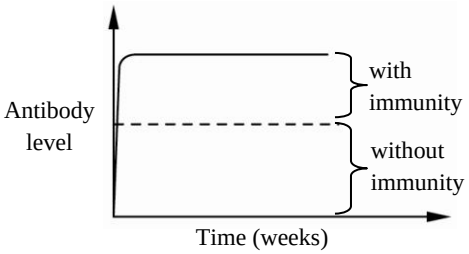
A man stepped on a nail and got hurt. A serum was injected into his body to protect him against the disease tetanus.

Which of the following graphs shows the changes in the level of antibodies against tetanus in his blood after the injection?

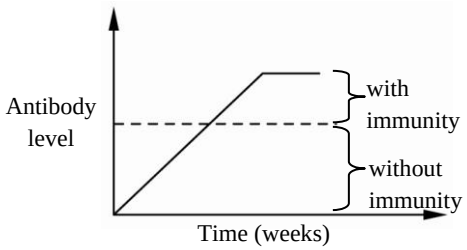
A.



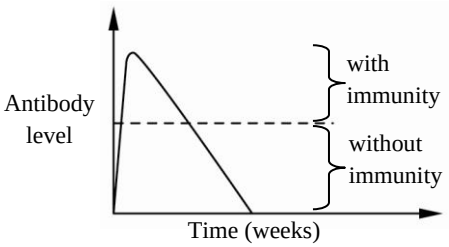
B.



C.



D.



□

##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0250000049|!|

Unlike the primary response, the secondary response is a process

(1) in which pathogens can be recognized by lymphocytes.

(2) in which more antibodies are produced.

(3) which lasts longer.

A. (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

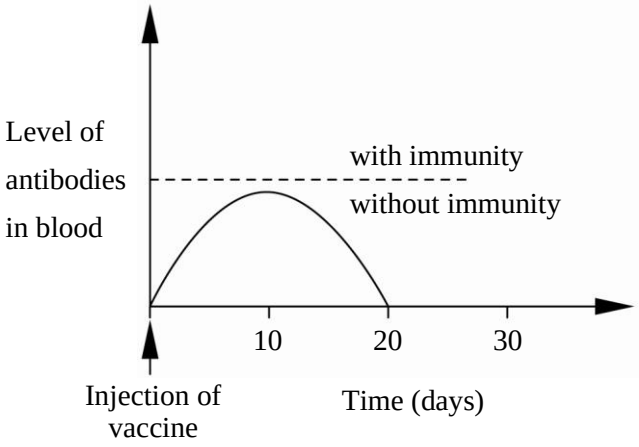
□

##C##

Skill assessed	Mark
Understanding basic concepts	1

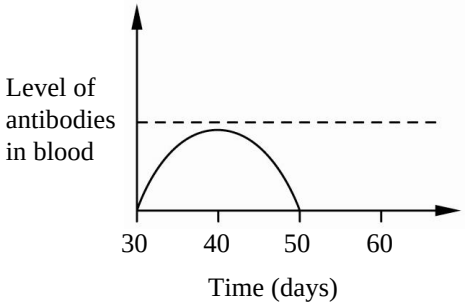
|!|BCEMB0250000050|!| ★

The graph below shows the changes in the level of antibodies against hepatitis B virus in the blood after an injection of vaccine against hepatitis B.

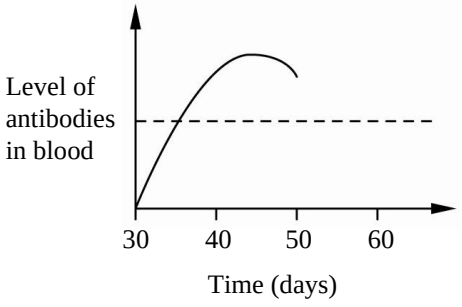


Which of the following graphs shows the changes in the level of antibodies in the blood after the second injection of the same vaccine on day 30?

A.

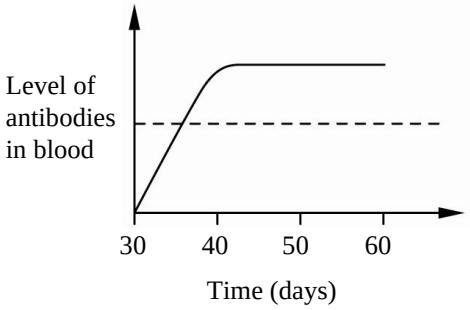


B.

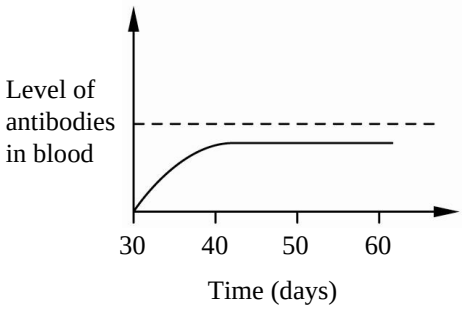


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C.



D.



##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0250000051|!|

Which of the following is/are the principles of vaccination for disease prevention?

(1) Antibodies are injected into the blood to provide immediate immunity.
(2) Subsequent exposure to the same antigen as in the vaccine will initiate a rapid production of a large amount of antibodies specific to that antigen.
(3) Memory cells memorize the antigen in the vaccine.

A. (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000052|!|

Which of the following comparisons between the primary response and the secondary response is ***incorrect***?

	Primary response	***Secondary response***
A.	slower	faster
B.	fewer antibodies are produced	more antibodies are produced
C.	short-term effect	long-term effect
D.	passive immunity is acquired	active immunity is acquired

##D##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000053|!|

Which of the following statements about different types of immunity is correct?

A. People can acquire active immunity artificially by vaccination.
B. People can acquire active immunity naturally by injection of antibodies.
C. Babies can acquire passive immunity artificially by being breast-fed.
D. People who have recovered from a disease can acquire passive immunity naturally.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000054|!|

Which of the following comparisons between active immunity and passive immunity is ***incorrect***?

	Active immunity	***Passive immunity***
A.	antibodies are produced by our own body	antibodies are not produced by our own body
B.	provides immediate immunity	does not provide immediate immunity
C.	long-lasting	short-lived
D.	can be acquired by vaccination	can be acquired by injection of serum

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000055|!|

What of the following account(s) for the difference between the primary response and the secondary response?

(1) During the first exposure to an antigen, memory cells specific for the antigen have been produced.
(2) During the second exposure to the antigen, memory cells divide and differentiate quickly into large numbers of plasma cells, killer T cells and memory cells.
(3) Antibodies specific to the antigens are produced only in the secondary response.

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A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000056|!|

Which of the following combinations correctly explain the signs and symptoms of the inflammatory response?

	Redness	***Heat***
A.	secretion of toxins by pathogens	secretion of antibodies by lymphocytes
B.	formation of a blood clot at the wound	increased phagocytosis at the wound
C.	formation of pus at the wound	accumulation of excess tissue fluid at the wound
D.	increased blood flow to the area around the wound	the increased blood flow raises the temperature of the wound

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0250000057|!| ★

Which of the following are the similarities between *Amoeba* and phagocytes?

(1) Both of them can engulf and digest other microorganisms.
(2) Both of them can replicate by binary fission.
(3) Both of them absorb oxygen by diffusion.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0250000058|!|

Which of the following comparisons between B cells and T cells is correct?

	B cells	***T cells***
A.	formed in the bone marrow	formed in the thymus gland
B.	can differentiate into memory cells	cannot differentiate into memory cells
C.	mature in the thymus gland	mature in the bone marrow
D.	can differentiate into plasma cells	cannot differentiate into plasma cells

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000059|!|

The following statements describe the stages of phagocytosis.

(1) Lysosomes fuse with the vacuole.
(2) The pathogen is surrounded by the pseudopodia of the phagocyte.
(3) The pathogen is enclosed in a vacuole.
(4) Lysozyme is released into the vacuole and digests the pathogen.

Which of the following arranges the stages of phagocytosis in the correct order?

A. (2) → (3) → (1) → (4)
B. (2) → (3) → (4) → (1)
C. (3) → (2) → (1) → (4)
D. (3) → (2) → (4) → (1)

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000060|!|

Which of the following ***incorrectly*** matches the defence mechanism with the substance involved?

	Defence mechanism	***Substance***
A.	inflammatory response	histamine
B.	blood clotting	fibrinogen
C.	phagocytosis	antibody
D.	immune response produced by T cells	lymphokine

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##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000061|!|

After the primary response, memory B cells remain in the blood for a long period of time. What is the function of memory B cells in the secondary exposure to the same pathogen?

A. produce a large amount of antibodies
B. engulf invading pathogens
C. differentiate into antibody-producing cells
D. differentiate into cells that can destroy cells infected by the same pathogen



##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0250000062|!|

Which of the following is/are caused by histamine?

(1) Dilation of blood capillaries near the wound
(2) Increased permeability of capillary walls
(3) Increased blood flow to the tissues around the wound

A. (1) only
B. (2) only
C. (2) and (3) only
D. (1), (2) and (3)



##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0250000063|!|

Which of the following cells can destroy cancer cells directly?

A. helper T cells
B. plasma cells
C. killer T cells
D. memory B cells

##C##

Skill assessed	Mark
Understanding basic concepts	1

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